A01 · FOUNDATION · C_6H_{14}

1. NUMBERED PARENT



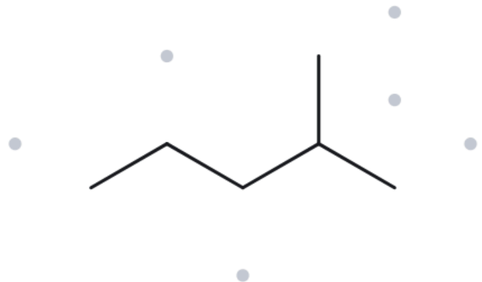
2. NAMING TRAIL

Parent: The longest continuous chain has 5 carbon atoms → pentane.
Numbering: Right to left gives 2; left to right gives 4. Choose 2 because it is the lower locant.
Substituents: One methyl group at carbon 2.

3 · MY NAME (WRITTEN FIRST)

2-methylpentane

4 · THE ALKANE CHECK

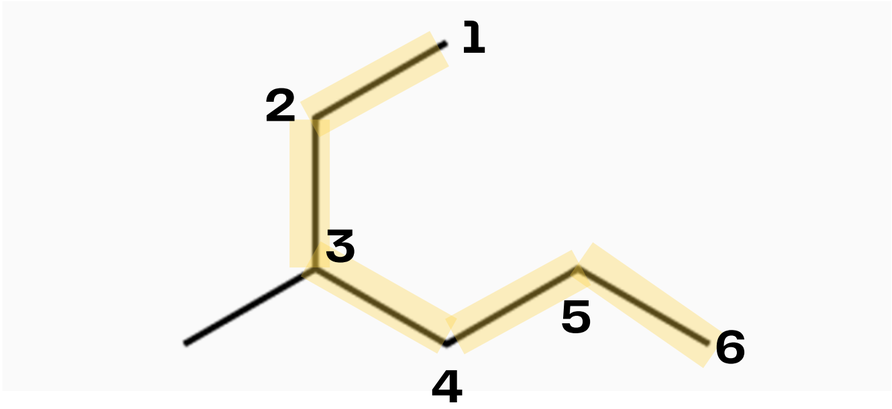


2-methylpentane
Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A02 · FOUNDATION · C_7H_{16}

1. NUMBERED PARENT



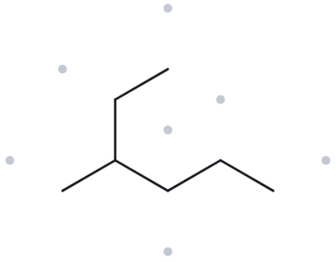
2. NAMING TRAIL

Parent: The longest continuous chain has 6 carbon atoms → hexane.
Numbering: Starting from the top end gives 3; starting from the right end gives 4. Choose 3 because it is the lower locant.
Substituents: One methyl group at carbon 3.

3 · MY NAME (WRITTEN FIRST)

3-methylhexane

4 · THE ALKANE CHECK



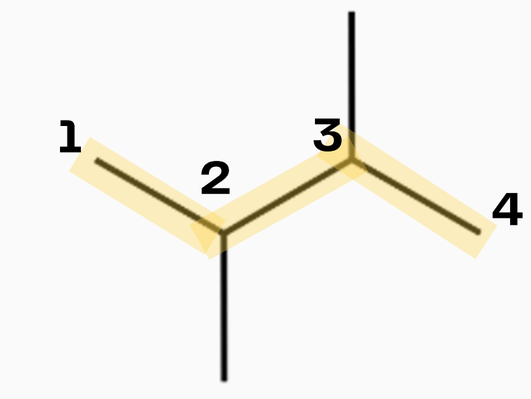
3-methylhexane

Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A03 · FOUNDATION · C_6H_{14}

1. NUMBERED PARENT



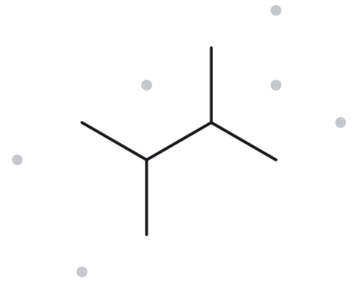
2. NAMING TRAIL

Parent: The longest continuous chain has 4 carbon atoms → butane.
Numbering: Both directions give 2,3, so either direction gives the same lowest locants.
Substituents: Two methyl groups at carbons 2 and 3 → 2,3-dimethyl.

3 · MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4 · THE ALKANE CHECK

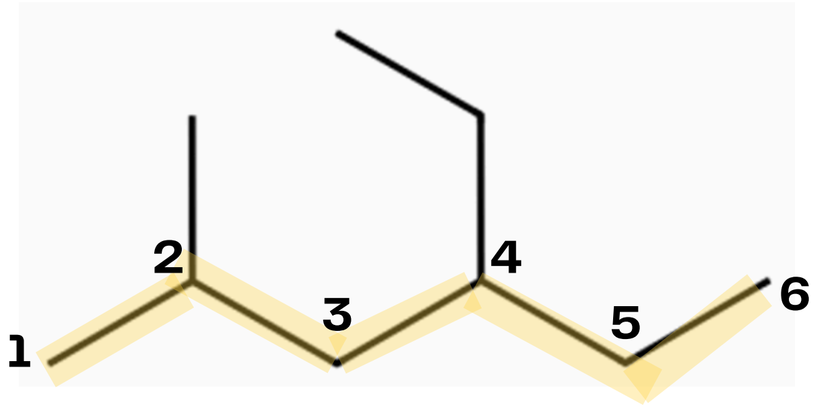


2,3-dimethylbutane
Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A04 · DEVELOPING · C_9H_{20}

1. NUMBERED PARENT



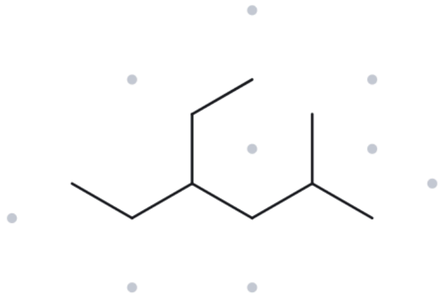
2. NAMING TRAIL

Parent: The longest continuous chain has 6 carbon atoms → hexane.
Numbering: Left to right gives 2,4; right to left gives 3,5. Choose 2,4 because 2 is lower than 3 at the first point of difference.
Substituents: Ethyl at carbon 4 and methyl at carbon 2. Cite ethyl before methyl in alphabetical order.

3 · MY NAME (WRITTEN FIRST)

4–ethyl–2–methylhexane

4 · THE ALKANE CHECK

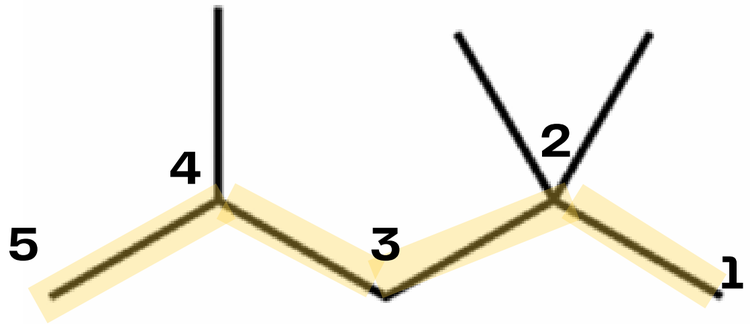


4-ethyl-2-methylhexane
Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A05 · DEVELOPING · C_8H_{18}

1. NUMBERED PARENT



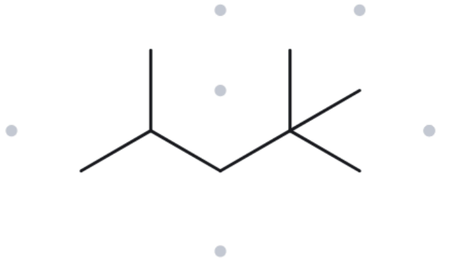
2. NAMING TRAIL

Parent: The longest continuous chain has 5 carbon atoms → pentane.
Numbering: Right to left gives 2,2,4; left to right gives 2,4,4. Choose 2,2,4 because the first locants tie, but the second locant 2 is lower than 4.
Substituents: Three methyl groups at carbons 2, 2 and 4 → 2,2,4–trimethyl.

3 · MY NAME (WRITTEN FIRST)

2,2,4–trimethylpentane

4 · THE ALKANE CHECK



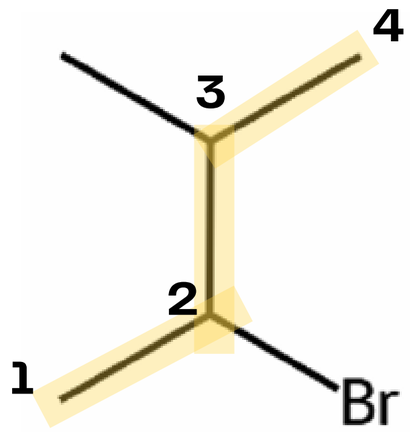
2,2,4-trimethylpentane

Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A06 · DEVELOPING · $C_5H_{11}Br$

1. NUMBERED PARENT



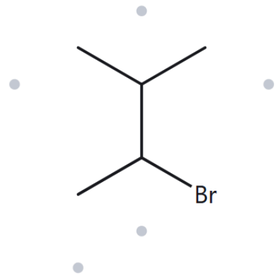
2. NAMING TRAIL

Parent: The longest continuous carbon chain has 4 carbon atoms → butane.
Numbering: Both directions give the locant set 2,3. Bromo comes before methyl alphabetically, so bromo receives the lower locant, 2.
Substituents: Bromo at carbon 2 and methyl at carbon 3, listed alphabetically.

3 · MY NAME (WRITTEN FIRST)

2–bromo–3–methylbutane

4 · THE ALKANE CHECK



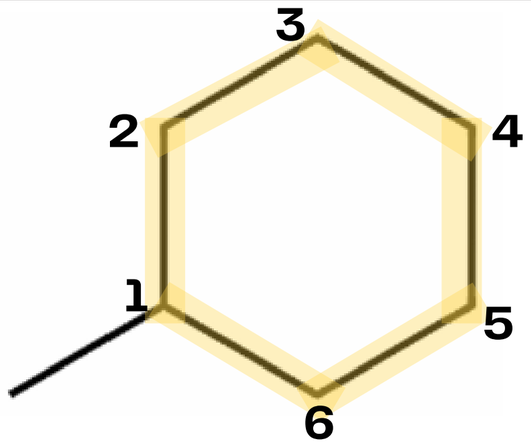
2-bromo-3-methylbutane

Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A07 · DEVELOPING · C_7H_{14}

1. NUMBERED PARENT



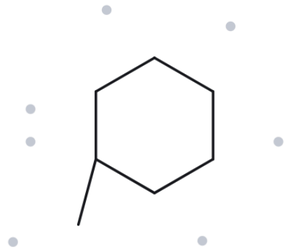
2. NAMING TRAIL

Parent: The parent is a 6-carbon ring → cyclohexane.
Numbering: Assign carbon `1` to the ring carbon bearing the methyl group. Either direction is equivalent. With only one substituent, the locant `1` is omitted from the name.
Substituents: One methyl group attached to the ring.

3 · MY NAME (WRITTEN FIRST)

methylcyclohexane

4 · THE ALKANE CHECK



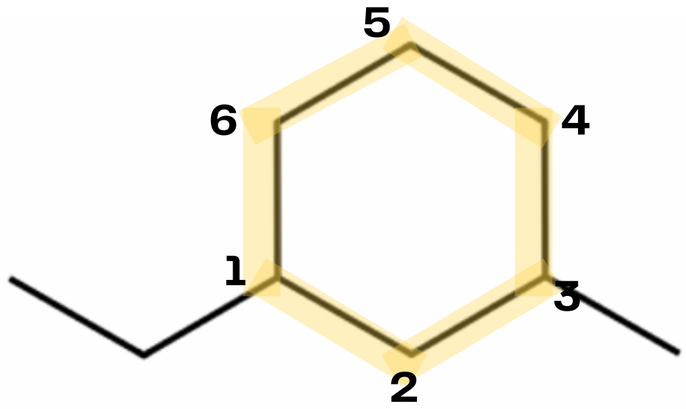
methylcyclohexane

Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A08 · CHALLENGE · C_9H_{18}

1. NUMBERED PARENT



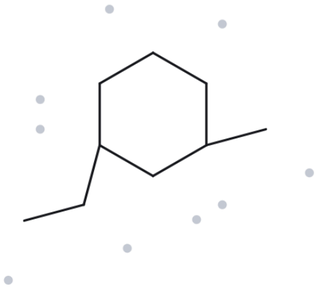
2. NAMING TRAIL

Parent: The parent is a 6-carbon ring → cyclohexane.
Numbering: Starting at the ethyl-bearing carbon gives 1,3 by the shorter route and 1,5 by the longer route. Choose 1,3. Starting at either substituent can give 1,3, so ethyl receives carbon 1 because it comes before methyl alphabetically.
Substituents: Ethyl at carbon 1 and methyl at carbon 3, listed alphabetically.

3 · MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4 · THE ALKANE CHECK



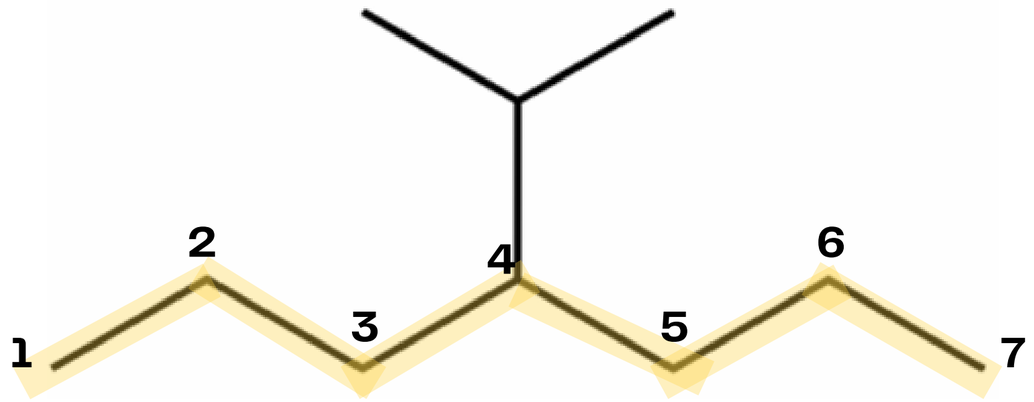
1-ethyl-3-methylcyclohexane

Match

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A09 · CHALLENGE · $C_{10}H_{22}$

1. NUMBERED PARENT



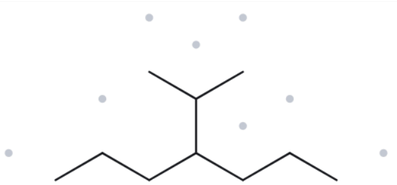
2. NAMING TRAIL

Parent: The longest continuous chain has 7 carbon atoms → heptane.
Numbering: Both directions place the substituent at carbon 4, so either direction gives the same locant.
Substituents: One isopropyl group at carbon 4. The group is attached through its middle carbon.

3 · MY NAME (WRITTEN FIRST)

4–isopropylheptane

4 · THE ALKANE CHECK



2-methyl-3-propylhexane

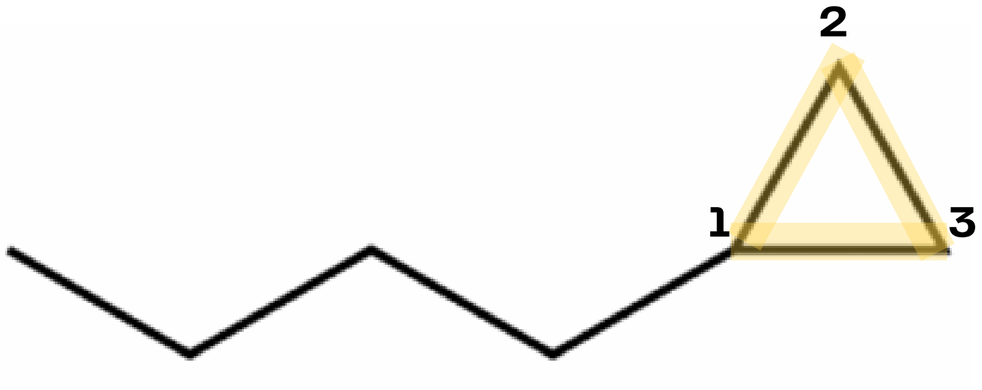
Mismatch

The correct name is 4–isopropylheptane. The longest continuous chain contains 7 carbon atoms, so the parent must be heptane. The tool incorrectly selected a 6–carbon parent chain.

☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A10 · CHALLENGE · C_7H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: The parent is the 3-carbon ring → cyclopropane. Under the naming convention specified in this activity, the ring is the parent even though the attached chain has more carbons.
Numbering: Assign carbon 1 to the ring carbon bearing the butyl group. Either direction is equivalent. With only one substituent, the locant 1 is omitted from the name.
Substituents: One straight-chain butyl group attached to the ring.

3 · MY NAME (WRITTEN FIRST)

butylcyclopropane

4 · THE ALKANE CHECK



butylcyclopropane
Match

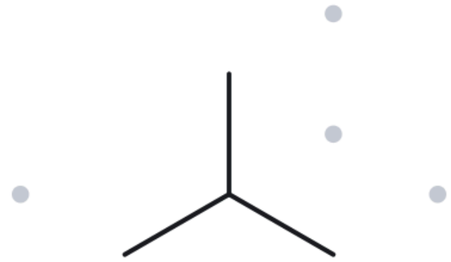
☑ parent confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

B01 · FOUNDATION · 2-methylpropane

1. NAME DECODED

Parent: Propane → a 3 – carbon chain.
Substituent: One methyl group at carbon 2.
Expected formula: 3 + 1 = 4 carbons.
For an open–chain alkane, C_nH_{2n+2} → C_4H_{10}.

2. BOND - LINE STRUCTURE



2-methylpropane

3. FORMULA CHECK

Counted from the drawing: C_4H_{10}
Open–chain alkane: C_nH_{2n+2}
For n = 4: H = 2(4) + 2 = 10.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 2–methylpropane
Match – the returned name matches the given name.

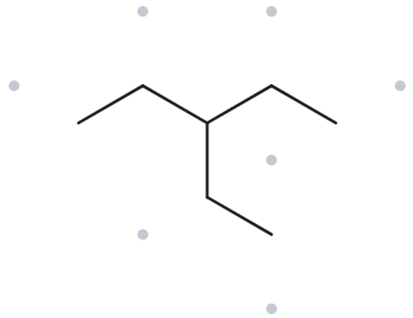
☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check recorded

B02 · FOUNDATION · 3-ethylpentane

1. NAME DECODED

Parent: Pentane → a 5-carbon chain.
Substituent: One ethyl group at carbon 3.
Expected formula: 5 + 2 = 7 carbons.
For an open-chain alkane, C_nH_{2n+2} → C_7H_{16}.

2. BOND - LINE STRUCTURE



3-ethylpentane

3. FORMULA CHECK

Counted from the drawing: C_7H_{16}
Open-chain alkane: C_nH_{2n+2}
For n = 7: H = 2(7) + 2 = 16.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 3-ethylpentane
Match – the returned name matches the given name.

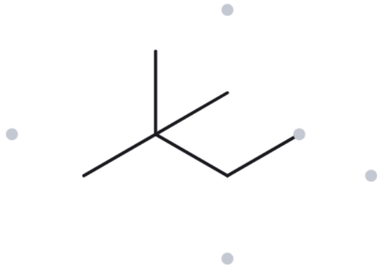
☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back-check recorded

B03 · FOUNDATION · 2,2-dimethylbutane

1. NAME DECODED

Parent: Butane → a 4–carbon chain.
Substituents: Two methyl groups, both at carbon 2.
Expected formula: 4 + 2 = 6 carbons.
For an open–chain alkane, C_nH_{2n+2} → C_6H_{14}.

2. BOND - LINE STRUCTURE



2,2-dimethylbutane

3. FORMULA CHECK

Counted from the drawing: C_6H_{14}
Open–chain alkane: C_nH_{2n+2}
For n = 6: H = 2(6) + 2 = 14.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 2,2–dimethylbutane
Match – the returned name matches the given name.

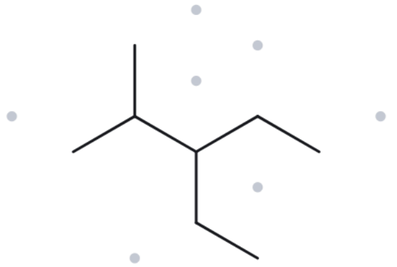
☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back–check recorded

B04 · DEVELOPING · 3-ethyl-2-methylpentane

1. NAME DECODED

Parent: Pentane → a 5-carbon chain.
Substituents: Ethyl at carbon 3 and methyl at carbon 2.
Expected formula: 5 + 2 + 1 = 8 carbons.
For an open-chain alkane, C_nH_{2n+2} → C_8H_{18}.

2. BOND - LINE STRUCTURE



3-ethyl-2-methylpentane

3. FORMULA CHECK

Counted from the drawing: C_8H_{18}
Open-chain alkane: C_nH_{2n+2}
For n = 8: H = 2(8) + 2 = 18.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 3-ethyl-2-methylpentane
Match – the returned name matches the given name.

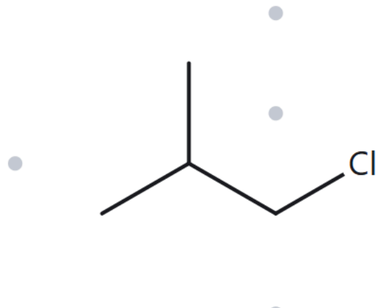
☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back-check recorded

B05 · DEVELOPING · 1-chloro-2-methylpropane

1. NAME DECODED

Parent: Propane → a 3-carbon chain.
Substituents: Chloro at carbon 1 and methyl at carbon 2.
Expected formula: 3 + 1 = 4 carbons.
The open-chain alkane formula is C_4H_{10}.
One Cl replaces one H → C_4H_9Cl.

2. BOND - LINE STRUCTURE



1-chloro-2-methylpropane

3. FORMULA CHECK

Counted from the drawing: C_4H_9Cl
For n = 4: C_nH_{2n+2} → C_4H_{10}.
Replacing one H with Cl gives C_4H_9Cl.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 1-chloro-2-methylpropane
Match – the returned name matches the given name.

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back-check recorded

B06 · DEVELOPING · 1,2-dimethylcyclopentane

1. NAME DECODED

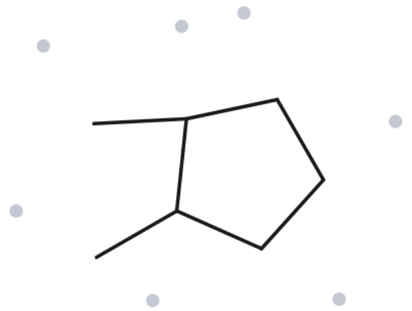
Parent: Cyclopentane → a 5-carbon ring.

Substituents: Two methyl groups at carbons 1 and 2.

Expected formula: 5 + 2 = 7 carbons.

For a compound with one saturated ring, C_nH_{2n} → C_7H_{14}.

2. BOND - LINE STRUCTURE



1,2-dimethylcyclopentane

3. FORMULA CHECK

Counted from the drawing: C_7H_{14}

One saturated ring: C_nH_{2n}

For n = 7: H = 2(7) = 14.

The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 1,2-dimethylcyclopentane

Match – the returned name matches the given name.

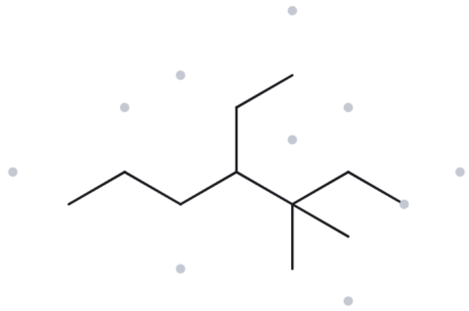
☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back-check recorded

B07 · DEVELOPING · 4-ethyl-3,3-dimethylheptane

1. NAME DECODED

Parent: Heptane → a 7-carbon chain.
Substituents: One ethyl group at carbon 4 and two methyl groups at carbon 3.
Expected formula: 7 + 2 + 2 = 11 carbons.
For an open-chain alkane, C_nH_{2n+2} → $C_{11}H_{24}$.

2. BOND - LINE STRUCTURE



4-ethyl-3,3-dimethylheptane

3. FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$
Open-chain alkane: C_nH_{2n+2}
For n = 11: H = 2(11) + 2 = 24.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 4-ethyl-3,3-dimethylheptane
Match – the returned name matches the given name.

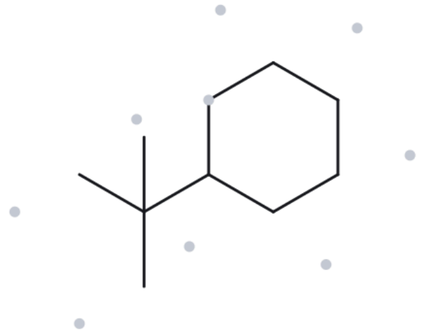
☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back-check recorded

B08 · CHALLENGE · tert-butylcyclohexane

1. NAME DECODED

Parent: Cyclohexane → a 6-carbon ring.
Substituent: One tert-butyl group, $-C(CH_3)_3$, attached to the ring through its central carbon.
Expected formula: 6 + 4 = 10 carbons.
For one saturated ring, C_nH_{2n} → $C_{10}H_{20}$.

2. BOND - LINE STRUCTURE



(1,1-dimethylethyl)cyclohexane

3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
One saturated ring: C_nH_{2n}
For n = 10: H = 2(10) = 20.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: (1,1-dimethylethyl)cyclohexane
Equivalent match
The substituent names tert-butyl and (1,1-dimethylethyl) describe the same $-C(CH_3)_3$ group. Both names represent the same compound.

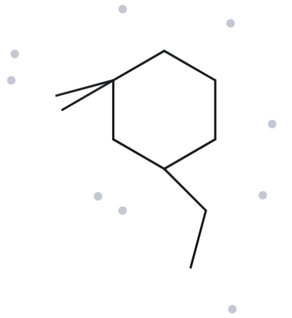
☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back-check recorded

B09 · CHALLENGE · 3-ethyl-1,1-dimethylcyclohexane

1. NAME DECODED

Parent: Cyclohexane → a 6-carbon ring.
Substituents: Two methyl groups at carbon 1 and one ethyl group at carbon 3.
Expected formula: 6 + 2 + 2 = 10 carbons.
For one saturated ring, C_nH_{2n} → $C_{10}H_{20}$.

2. BOND - LINE STRUCTURE



3-ethyl-1,1-dimethylcyclohexane

3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
One saturated ring: C_nH_{2n}
For n = 10: H = 2(10) = 20.
The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 3-ethyl-1,1-dimethylcyclohexane
Match – the returned name matches the given name.

☑ Parent length matches ☑ Substituent positions match ☑ Formula agrees ☑ Back-check recorded

B10 · CHALLENGE · 2-ethylbutane

deliberately incorrect name

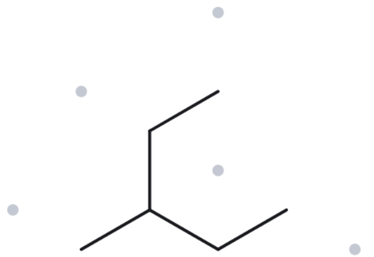
1. NAME DECODED

As written: Butane → a 4-carbon chain, with an ethyl group at carbon 2.

Expected formula: 4 + 2 = 6 carbons.

For an open-chain alkane, C_nH_{2n+2} → C_6H_{14}.

2. BOND - LINE STRUCTURE



3-methylpentane

3. FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open-chain alkane: C_nH_{2n+2}

For n = 6: H = 2(6) + 2 = 14.

The counted and expected formulas agree.

4. BACK - CHECK

Returned name: 3-methylpentane

Mismatch – the given name is incorrect.

The given name 2-ethylbutane uses a 4-carbon parent even though the longest continuous chain has 5 carbon atoms. The correct parent is pentane, with a methyl group at carbon 3. Therefore, the correct name is 3-methylpentane.

☑ **Parent length matches** ☑ **Substituent positions match** ☑ **Formula agrees** ☑ **Back-check recorded**